25901224 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECT200 - PYTHON PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which data type is used to store true/false values? [K] / [CO1]

   a). bool b). int c). str d). float

2. Which statement is used to check multiple conditions in Python? [U] / [CO2]

   a). if block b). else

   c). case d). elif

3. Which statement is used to import a package? [K] / [CO3]

   a). include package

   b). load package

   c). add package

   d). import package

4. Which operator is used for floor division? [U] / [CO1]

a). / b). % c). // d). **

5. What is the purpose of the exit() function in Python? [K] / [CO2]

a). To terminate the program execution

b). To skip the current loop iteration

c). To pause the program

d). To restart the loop

6. All built-in exceptions in Python are derived from which base class? [A] / [CO3]

a). Exception b).RuntimeError c).BaseException d). SystemError

7. Python can be executed in [K] / [CO1]

a). Only IDE mode b). Only command line mode

c). Both command line and IDE d). Compiler mode

8. Which statement is used to exit a while loop immediately? [U] / [CO2]

a). stop b). exit

c). break d). continue

9. What does the character set [a-z] match? [K] / [CO3]

a). Any sequence of lowercase letters b). The literal string 'a-z'

c). Any single lowercase letter from a to z d). Any letter,regardless of case

10. Which function in Matplotlib is used to plot a simple line graph? [A] / [CO3]

a). line() b). plot() c). graph() d). bar()

[1x10 = 10]

Part B

Short Answer Question

Answer any **4** questions

Each question carries **5** marks

11. Explain dictionaries in python and discuss any two common dictionary methods [U] / [CO1]

12. Describe the for loop in Python. Write a program to print numbers from 1 to 10 [K] / [CO2]

13. Explain the "Arguments of an Exception" with an example. [A] / [CO3]

14. Describe accessing characters in a string with examples. [K] / [CO1]

15. Distinguish between a Histogram and a Bar Chart based on their structure and data usage. [A] / [CO3]

16. Write short notes on common string methods. [U] / [CO1]

[5x4 = 20]

Part C

Essay Question

Answer any **2** questions

Each question carries **10** marks

17. Explain variables in Python with examples from electronics applications. [U] / [CO1]

18. Explain Loop Control Flow in Python. Describe for loop and while loop along with loop control statements (break, continue, pass) using examples [U] / [CO2]

19. Explain function calling, passing arguments, and the return statement in Python with suitable examples [U] / [CO2]

20. Describe the complete workflow of using regular expressions in Python. [U] / [CO3]

[10x2 = 20]